



Rendering

Rendering is the process of generating a 2D Image from a 3D Scene by calculating the color of every pixel in the final image based on geometry, materials, lighting, cameras, and effects.

Rendering combines:

- Geometry: 3D models (spheres, cylinders, triangles...)
- Materials: Color, texture, shininess, transparency
- Lighting: Lights, shadows, brightness, color
- Camera: Position, angle, field of view, lens effects
- Effects: Fog, reflections, motion blur, depth of field

The Renderer's job is to simulate how light interacts with objects in the scene and what reaches the virtual camera.

Main Approaches

There are two main approaches to Rendering:

- Rasterization
 - Projects geometry onto the screen, then shades pixels
 - Used for games, UI, and VR
- Ray Tracing
 - Shoots rays from the camera into the scene, sees what they hit
 - Used for offline movies and visual effects, or for real-time with RTX

The Rendering Pipeline

Rendering usually follows these high-level steps:

1. Scene Setup: Load models, lights, cameras, and materials
2. Visibility Determination: Decide which objects are visible to the camera
3. Light Transport Simulation: Calculate how light bounces around the scene
4. Shading: Determine the color of each visible surface point
5. Post-Processing: Apply filters, tone mapping, anti-aliasing
6. Display: Output final pixels to screen or image file